Energy Methods and its Applications to Uniqueness of Initial Boundary Value Problems

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Course: Partial Differential Equations

Abstract

This report presents a self-contained study of the *energy method* for Initial Boundary Value Problems (IBVPs) in partial differential equations. It mainly focus on proving *uniqueness* of solutions. We first introduce the general framework of IBVPs and motivate the method by discussing Sharp Hadamard's notion of well-posedness. The necessary mathematical tools — the L^2 norm, integration by parts, and Gronwall's inequality — are developed carefully. The energy method is then applied to the heat equation (Dirichlet and Neumann boundary conditions) and to the wave equation (including the damped case), each time yielding clean proofs of uniqueness. Two fully worked examples with every algebraic step are included. The report concludes with a discussion of the advantages and limitations of the energy approach.

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1. Introduction

Partial differential equations (PDEs) arise throughout the natural sciences and engineering: the heat equation models diffusion of temperature, the wave equation governs vibrating strings and sound, and the Laplace/Poisson equations describe electrostatic potentials and steady-state temperatures [1, 2, 8]. In each case, a physical model is only meaningful if the associated mathematical problem has a solution that is *unique* and *stable* with respect to the given data.

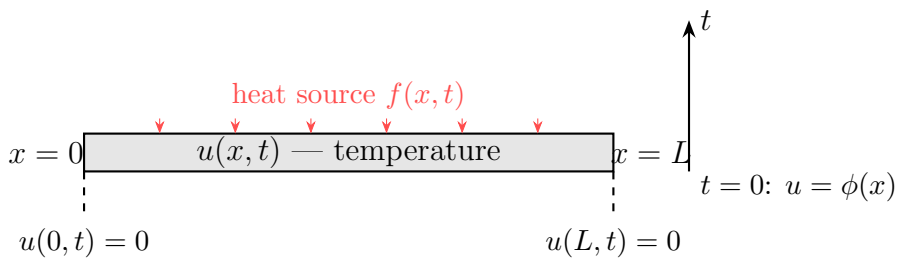

Figure 1: A heated rod of length L with Dirichlet (zero-temperature) boundary conditions and initial temperature profile $\phi(x)$. The IBVP asks for the temperature $u(x,t)$ for all $t>0$.

1.1. Initial Boundary Value Problems

An **Initial Boundary Value Problem (IBVP)** consists of three parts:

(i) A PDE that $u(x,t)$ must satisfy in the interior of the domain for $t>0$,

(ii) **Boundary conditions (BCs)** specifying the behaviour of u on the boundary of the spatial domain for all $t \geq 0$,

(iii) An **initial condition (IC)** specifying u (and possibly its time derivatives) at $t=0$.

In these notes the spatial domain is the interval $(0,L)$, so the general IBVP reads

$$\mathcal{L}[u] = f(x,t), \qquad x \in (0,L),\ t>0, \tag{1}$$

$$u(x,0) = \phi(x), \qquad x \in [0,L], \tag{2}$$

$$\text{Boundary conditions at } x=0 \text{ and } x=L, \quad t \geq 0. \tag{3}$$

The most common boundary conditions are:

- **Dirichlet:** $u(0,t)=g_1(t)$, $u(L,t)=g_2(t)$,
- **Neumann:** $u_x(0,t)=h_1(t)$, $u_x(L,t)=h_2(t)$,
- **Robin/Mixed:** $u_x+\alpha u=0$ at the boundary .

1.2. Well-Posedness and the Role of Uniqueness

Hadamard [3] (1902) proposed that a PDE problem is *well-posed* if its solution: (i) **exists**, (ii) is **unique**, and (iii) **depends continuously on the data** (stability). Physical models that fail any of these conditions are unreliable; small measurement errors in the data could produce arbitrarily large errors in the prediction.

The *energy method* is a technique that addresses the uniqueness and stability questions *without* needing to find the explicit solution. It assigns a non-negative scalar quantity $E(t)$ — the *energy* — to the solution and tracks how $E(t)$ evolves in time. If the energy of the *difference* of two assumed solutions is shown to be identically zero, the two solutions must coincide.

2. Mathematical Background

2.1. The L^2 Norm and Inner Product

Definition 2.1 (L^2 Norm)**.** For a function $f : (0, L) \to \mathbb{R}$, the L^2 norm is defined by

$$\|f\|^2 \;=\; \|f\|^2_{L^2(0,L)} \;=\; \int_0^L [f(x)]^2\, dx. \tag{4}$$

The L^2 norm captures the *global* size of f over the whole interval, in contrast with the maximum norm which only measures the largest pointwise value. It arises naturally in PDEs because:

- For the heat equation, $\int u^2\, dx$ is related to total thermal energy.
- For the wave equation, $\int (u_t^2 + u_x^2)\, dx$ represents the sum of kinetic and potential energy.
- Integration and differentiation interact cleanly via integration by parts [4].

Definition 2.2 (L^2 Inner Product)**.** For two functions f, g on $(0, L)$:

$$\langle f, g\rangle \;=\; \int_0^L f(x)\, g(x)\, dx. \tag{5}$$

Note that $\langle f, f\rangle = \|f\|^2$ [6].

2.2. Integration by Parts

The single most-used algebraic tool in the energy method is integration by parts (IBP), which follows directly from the product rule.

Lemma 2.3 (Integration by Parts)**.** *If f and g are differentiable on $[0, L]$, then*

$$\int_0^L f'(x)\, g(x)\, dx \;=\; \big[f(x)g(x)\big]_0^L \;-\; \int_0^L f(x)\, g'(x)\, dx. \tag{6}$$

Setting $f = u_x$ and $g = u$ in (6) yields the key identity

$$\int_0^L u_{xx}\, u\, dx \;=\; \big[u_x\, u\big]_0^L \;-\; \int_0^L u_x^2\, dx. \tag{7}$$

Under **homogeneous Dirichlet BCs** ($u(0,t) = u(L,t) = 0$), the boundary term vanishes, giving

$$\int_0^L u_{xx}\, u\, dx \;=\; -\int_0^L u_x^2\, dx \;=\; -\,\|u_x\|^2\,. \tag{8}$$

This identity converts the second derivative u_{xx} into the first derivative u_x, which is the key step that makes the energy method work.

2.3. Differentiation of a Squared Norm

A simple but fundamental identity governs how the L^2 norm of a time-dependent function evolves:

$$\frac{d}{dt}\int_0^L [u(x,t)]^2\, dx \;=\; \int_0^L 2\, u\, u_t\, dx \;=\; 2\langle u, u_t\rangle, \tag{9}$$

obtained by differentiating under the integral sign. In short, $\dfrac{d}{dt}\,\|u\|^2 = 2\langle u, u_t\rangle$.

2.4. Gronwall's Inequality

Gronwall's inequality [5] converts a differential or integral inequality into an explicit upper bound — it is the backbone of every energy estimate.

Theorem 2.4 (Gronwall's Inequality — Integral Form)**.** *Let $E(t) \geq 0$ be continuous and satisfy*

$$E(t) \;\leq\; A + B\int_0^t E(s)\, ds, \qquad t \geq 0, \tag{10}$$

for constants $A \geq 0$, $B \geq 0$. Then

$$E(t) \;\leq\; A\, e^{Bt} \qquad \textit{for all } t \geq 0. \tag{11}$$

Proof. Define $H(t) = A + B\int_0^t E(s)\, ds$. Then $H(0) = A$, $H'(t) = BE(t) \leq BH(t)$. Dividing by $H(t) > 0$ and integrating from 0 to t:

$$\ln H(t) - \ln A \leq Bt \implies H(t) \leq Ae^{Bt}.$$

Since $E(t) \leq H(t)$, the result follows. □

Corollary 2.5 (Gronwall's Inequality — Differential Form)**.** *If $E(t) \geq 0$ is differentiable and $E'(t) \leq C\, E(t)$ for a constant $C \in \mathbb{R}$, then*

$$E(t) \;\leq\; E(0)\, e^{Ct}. \tag{12}$$

In particular, if $C \leq 0$, then $E(t) \leq E(0)$ for all $t > 0$ — the energy does not grow.

Remark 2.6. The Poincaré inequality is also used in energy decay estimates:

$$\|v\|^2 \;\leq\; \frac{L^2}{\pi^2}\,\|v_x\|^2 \tag{13}$$

for any $v \in H_0^1(0,L)$ [7] (i.e., $v(0) = v(L) = 0$). This allows one to bound the L^2 norm of v by the L^2 norm of its derivative.

3. The Energy Method: Main Theory

3.1. The Energy Functional

In the energy method, we assign a scalar quantity $E(t) \geq 0$ to the solution $u(\cdot, t)$, called the *energy functional*. The choice depends on the equation:

- **Heat equation** (first order in t): $E(t) = \frac{1}{2} \|u(\cdot, t)\|^2 = \frac{1}{2} \int_0^L u^2 \, dx$.
- **Wave equation** (second order in t): $E(t) = \frac{1}{2} \int_0^L (u_t^2 + c^2 u_x^2) \, dx$, which is kinetic + potential energy.

3.2. The Two-Step Strategy

Energy Method — Two-Step Strategy

Step 1 (Energy identity). Multiply the PDE by u (for the heat equation) or by u_t (for the wave equation). Integrate over $x \in (0, L)$, apply IBP, and use the boundary conditions to eliminate boundary terms. This yields a relation for $E'(t)$.

Step 2 (Gronwall / integration). The result from Step 1 is typically an inequality $E'(t) \leq CE(t) + G(t)$. Apply Gronwall's inequality to obtain an explicit bound on $E(t)$.

3.3. How Uniqueness Follows

Suppose u_1 and u_2 both solve the same IBVP. Define the difference $w = u_1 - u_2$. By linearity of the PDE operator:

- w satisfies the *homogeneous* PDE (source terms cancel),
- w satisfies *homogeneous* boundary conditions ($w = 0$ on the boundary),
- w satisfies *zero* initial condition ($w(x, 0) = 0$).

Apply the energy method to w. Define $E_w(t) = \frac{1}{2} \|w\|^2 \geq 0$. If the energy identity gives $E_w'(t) \leq 0$, then since E_w is non-increasing and

$$E_w(0) = \frac{1}{2} \int_0^L [w(x, 0)]^2 \, dx = 0,$$

it follows that $0 \leq E_w(t) \leq E_w(0) = 0$, i.e. $E_w(t) = 0$ for all $t \geq 0$, which forces $w \equiv 0$, hence $u_1 = u_2$.

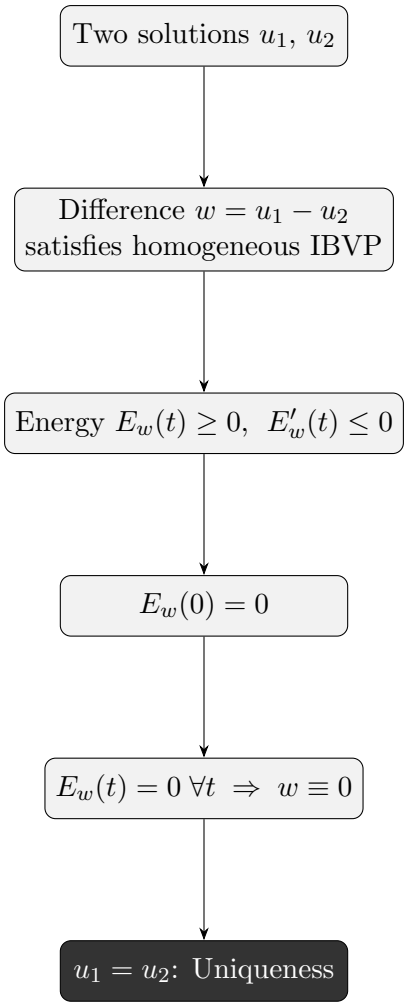

Figure 2: Flowchart of the uniqueness argument via the energy method.

3.4. How Stability Follows

Let u_1 solve the IBVP with data (ϕ_1, f_1) and u_2 with data (ϕ_2, f_2). Set $w = u_1 - u_2$ and $F = f_1 - f_2$. The energy method applied to w yields a bound of the form

$$E_w(t) \ \leq\ C(T)\Big(\|\phi_1 - \phi_2\|^2 + \int_0^t \|F(\cdot, s)\|^2\ ds\Big).$$

If $\phi_1 \to \phi_2$ and $f_1 \to f_2$, then $E_w(t) \to 0$ — small changes in data produce small changes in the solution. This is continuous dependence on data, the stability aspect of well-posedness.

4. Applications

4.1. Application 1: The Heat Equation (Dirichlet BCs)

Problem Setup

Heat Equation IBVP

$$u_t = k\,u_{xx} + f(x,t), \qquad x \in (0,L),\ t > 0, \tag{14}$$
$$u(x,0) = \phi(x), \qquad x \in [0,L], \tag{15}$$
$$u(0,t) = 0, \quad u(L,t) = 0, \qquad t > 0. \tag{16}$$

Here $k > 0$ is the thermal diffusivity, f is a heat source/sink, and ϕ is the initial temperature profile.

Deriving the Energy Identity

Define $E(t) = \frac{1}{2}\|u(\cdot,t)\|^2$. Differentiating and substituting the PDE:

$$E'(t) = \int_0^L u\,u_t\,dx = \int_0^L u\,(k\,u_{xx} + f)\,dx = k\int_0^L u\,u_{xx}\,dx + \langle u, f\rangle.$$

Applying (8) (boundary terms vanish by (16)):

$$\boxed{E'(t) = -k\,\|u_x\|^2 + \langle u, f\rangle.} \tag{17}$$

The term $-k\,\|u_x\|^2 \le 0$ shows that diffusion always *dissipates* energy; the source term $\langle u, f\rangle$ can add or remove energy.

Uniqueness for the Heat Equation

Theorem 4.1 (Uniqueness — Heat Equation)**.** *The IBVP* (14)–(16) *has at most one solution.*

Proof. Let u_1, u_2 solve (14)–(16) and set $w = u_1 - u_2$. Then w satisfies $w_t = k\,w_{xx}$ with $w(x,0) = 0$ and homogeneous Dirichlet BCs. Define $E_w(t) = \frac{1}{2}\|w\|^2 \ge 0$. By (17) with $f = 0$:

$$E_w'(t) = -k\,\|w_x\|^2 \le 0.$$

So E_w is non-increasing. Since $E_w(0) = \frac{1}{2}\int_0^L [w(x,0)]^2\,dx = 0$,

$$0 \le E_w(t) \le E_w(0) = 0 \implies E_w(t) = 0 \ \ \forall\, t \ge 0.$$

Therefore $\int_0^L w^2\,dx = 0$, forcing $w \equiv 0$, i.e. $u_1 = u_2$. □

Energy Decay (Homogeneous Case $f = 0$)

Using the Poincaré inequality (13) in (17) (with $f = 0$):

$$\|u_x\|^2 \ge \frac{\pi^2}{L^2}\,\|u\|^2 = \frac{2\pi^2}{L^2}E(t).$$

Hence $E'(t) \le -\frac{2k\pi^2}{L^2}E(t)$, and by Gronwall's inequality:

$$E(t) \ \le\ E(0)\,e^{-2k\pi^2 t/L^2}. \tag{18}$$

The energy decays *exponentially* to zero — the rod eventually cools to zero temperature, as expected physically.

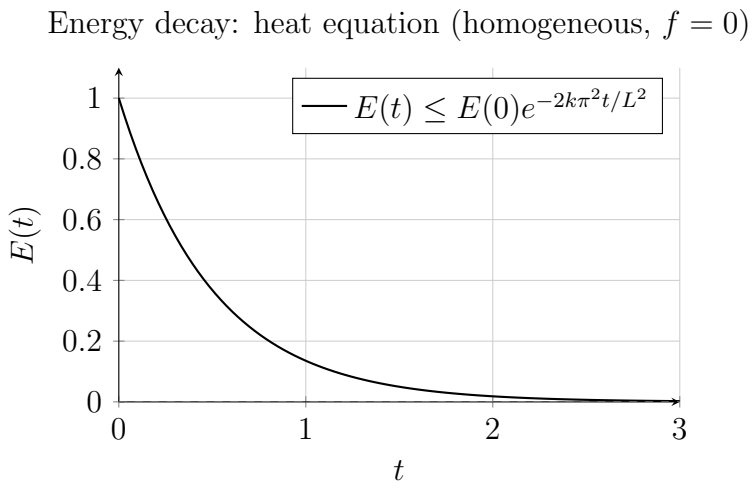

Figure 3: Exponential decay of the energy $E(t)$ for the homogeneous heat equation with Dirichlet boundary conditions. The constant in the exponent depends on k and L.

4.2. Application 2: Heat Equation with Neumann Boundary Conditions

Neumann IBVP (Insulated Ends)

$$u_t = k\, u_{xx}, \qquad x \in (0, L),\ t > 0, \tag{19}$$

$$u(x, 0) = \phi(x), \qquad x \in [0, L], \tag{20}$$

$$u_x(0, t) = 0, \quad u_x(L, t) = 0, \qquad t > 0. \tag{21}$$

These BCs model a rod whose ends are thermally insulated (no heat flux). Integration by parts now gives

$$\int_0^L u\, u_{xx}\, dx = \big[u\, u_x\big]_0^L - \|u_x\|^2 = u(L)u_x(L) - u(0)u_x(0) - \|u_x\|^2 = -\,\|u_x\|^2\,,$$

since $u_x(0, t) = u_x(L, t) = 0$. The energy identity is the same as (17) (with $f = 0$), so uniqueness follows by the same argument. In addition, integrating the PDE over $(0, L)$ shows that the total heat is conserved:

$$\frac{d}{dt}\int_0^L u\, dx = k\int_0^L u_{xx}\, dx = k\big[u_x\big]_0^L = 0.$$

4.3. Application 3: The Wave Equation

Problem Setup

Wave Equation IBVP

$$u_{tt} = c^2 u_{xx} + f(x,t), \qquad x \in (0,L),\ t > 0, \tag{22}$$
$$u(x,0) = \phi(x), \quad u_t(x,0) = \psi(x), \qquad x \in [0,L], \tag{23}$$
$$u(0,t) = 0, \quad u(L,t) = 0, \qquad t > 0. \tag{24}$$

Here $c > 0$ is the wave speed, ϕ is the initial displacement, and ψ is the initial velocity.

Choosing the Right Energy

For a second-order-in-time equation we include both kinetic and potential energy:

$$E(t) = \frac{1}{2}\int_0^L \left(u_t^2 + c^2 u_x^2\right) dx. \tag{25}$$

Note $E(t) \geq 0$ and $E(t) = 0 \iff u_t \equiv 0$ and $u_x \equiv 0$, which with the BCs implies $u \equiv 0$.

Deriving the Energy Identity

Multiply the PDE (22) by u_t and integrate over $(0,L)$:

$$E'(t) = \int_0^L (u_t u_{tt} + c^2 u_x u_{xt})\, dx.$$

Apply IBP to the term $c^2 \int_0^L u_x u_{xt}\, dx$ (boundary terms vanish since $u_t(0,t) = u_t(L,t) = 0$ from the Dirichlet BCs):

$$c^2 \int_0^L u_x u_{xt}\, dx = -c^2 \int_0^L u_{xx} u_t\, dx.$$

Hence

$$E'(t) = \int_0^L u_t\,(u_{tt} - c^2 u_{xx})\, dx = \int_0^L u_t\, f\, dx,$$

where we used $u_{tt} - c^2 u_{xx} = f$. This gives the energy identity:

$$\boxed{E'(t) = \langle u_t, f\rangle.} \tag{26}$$

When $f = 0$: $E'(t) = 0$, so $E(t) = E(0)$ — the wave equation *conserves* energy, in sharp contrast to the heat equation.

Uniqueness for the Wave Equation

Theorem 4.2 (Uniqueness — Wave Equation)**.** *The IBVP* (22)–(24) *has at most one solution.*

Proof. Let u_1, u_2 solve the IBVP and set $w = u_1 - u_2$. Then w satisfies $w_{tt} = c^2 w_{xx}$ with $w(x,0) = 0$, $w_t(x,0) = 0$, and homogeneous BCs. Define $E_w(t) = \frac{1}{2}\int_0^L (w_t^2 + c^2 w_x^2)\, dx \geq 0$. By (26) with $f = 0$: $E_w'(t) = 0$, so $E_w(t) = E_w(0)$ for all t. Computing $E_w(0)$:

$$E_w(0) = \frac{1}{2}\int_0^L \left([w_t(x,0)]^2 + c^2[w_x(x,0)]^2\right) dx = 0,$$

since $w_t(x,0) = 0$ and $w_x(x,0) = \frac{\partial}{\partial x}[w(x,0)] = 0$. Therefore $E_w(t) = 0$, forcing $w_t \equiv 0$ and $w_x \equiv 0$. So w is constant in x and t; since $w(x,0) = 0$, we get $w \equiv 0 \implies u_1 = u_2$. $\square$

5. Worked Examples

5.1. Example 1: Heat Equation

Example 5.1. Consider the IBVP on $(0, \pi)$:

$$\begin{aligned} u_t &= u_{xx} + u, \quad x \in (0, \pi),\ t > 0, \\ u(x, 0) &= \sin x, \quad x \in [0, \pi], \\ u(0, t) &= 0, \quad u(\pi, t) = 0. \end{aligned}$$

(a) Find the energy equation. **(b)** Determine whether the energy grows or decays. **(c)** Prove uniqueness using the energy method.

Solution.

Part (a). Take $k = 1$, $f = u$, $L = \pi$. Define $E(t) = \frac{1}{2} \|u\|^2$. Applying (17) with $f = u$:

$$E'(t) = -\|u_x\|^2 + \langle u, u \rangle = -\|u_x\|^2 + \|u\|^2 .$$

Part (b). By the Poincaré inequality (13) with $L = \pi$:

$$\|u_x\|^2 \geq \frac{\pi^2}{\pi^2} \|u\|^2 = \|u\|^2 .$$

Therefore:

$$E'(t) = -\|u_x\|^2 + \|u\|^2 \leq -\|u\|^2 + \|u\|^2 = 0.$$

Hence $E'(t) \leq 0$: the energy is *non-increasing*. One can verify with the exact solution $u(x, t) = \sin x$ (since $(\sin x)_t = 0 = (\sin x)_{xx} + \sin x$): the energy is constant $E(t) = \pi/4$, consistent with $E'(t) \leq 0$.

Part (c). For uniqueness, let $w = u_1 - u_2$ satisfy $w_t = w_{xx} + w$, $w(x, 0) = 0$, homogeneous BCs. From Part (b), $E_w'(t) \leq 0$ and $E_w(0) = 0$, so $E_w(t) = 0$ and $w \equiv 0$. Uniqueness follows.

5.2. Example 2:Damped Wave Equation

Example 5.2. Consider the damped wave equation:

$$\begin{aligned} u_{tt} + 2\gamma u_t &= c^2 u_{xx}, \quad x \in (0, L),\ t > 0, \\ u(x, 0) &= \phi(x), \quad u_t(x, 0) = \psi(x), \\ u(0, t) &= 0, \quad u(L, t) = 0. \end{aligned}$$

Here $\gamma > 0$ is the damping coefficient. **(a)** Derive the energy equation. **(b)** Show the energy is non-increasing. **(c)** Prove uniqueness.

Solution.

Part (a). Use $E(t) = \frac{1}{2} \int_0^L (u_t^2 + c^2 u_x^2)\, dx$. Multiplying the PDE by u_t and integrating (same IBP steps as Section 4):

$$E'(t) = \int_0^L u_t \left(u_{tt} - c^2 u_{xx}\right) dx.$$

The PDE gives $u_{tt} - c^2 u_{xx} = -2\gamma u_t$. Therefore:

$$E'(t) = -2\gamma \|u_t\|^2 \leq 0. \tag{27}$$

Part (b). Since $\gamma > 0$ and $\|u_t\|^2 \geq 0$, we have $E'(t) \leq 0$, so $E(t) \leq E(0)$ for all t. The damping term dissipates energy — the string gradually comes to rest.

Part (c). For uniqueness, let $w = u_1 - u_2$ satisfy the homogeneous damped wave equation with zero initial data. Then $E_w'(t) = -2\gamma \|w_t\|^2 \leq 0$ and $E_w(0) = 0$, so $E_w(t) = 0$ and $w \equiv 0$. □

Energy comparison: heat (decays) vs wave (conserved) vs damped wave

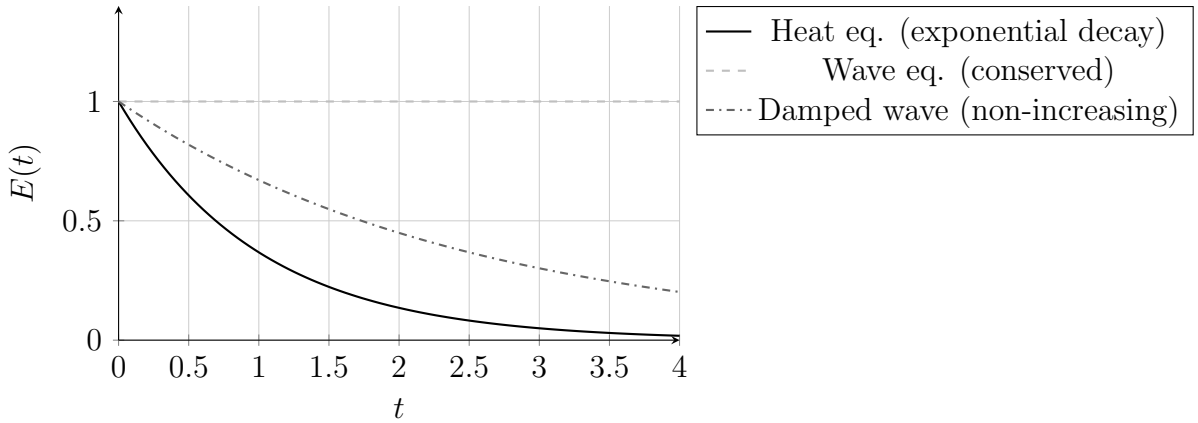

Figure 4: Qualitative energy behaviour for the three equations studied. The heat equation dissipates energy exponentially; the undamped wave equation conserves it; the damped wave equation dissipates it at a rate controlled by the damping coefficient γ.

6. Discussion

6.1. Strengths of the Energy Method

1. **No explicit solution required.** The method establishes uniqueness purely from the structure of the equation and the boundary conditions, without ever computing $u(x,t)$ in closed form.

2. **Flexibility.** The same two-step strategy (derive an energy identity, apply Gronwall) works for linear and many nonlinear PDEs, in one or more space dimensions.

3. **Quantitative stability.** The energy estimate directly yields an explicit bound on how much the solution can change when the data changes — a quantitative stability result.

4. **Physical insight.** The choice of energy functional mirrors physical energy, giving the mathematics a direct connection to the underlying physics (thermal energy, mechanical energy, etc.).

6.2. Limitations

1. **Only L^2 information.** Energy estimates control the L^2 norm of the solution. They do not directly give pointwise information (e.g. the maximum of u).

2. **Linear problems.** For nonlinear PDEs, deriving a useful energy identity can be significantly harder. Additional terms may appear that cannot be absorbed into the energy.

3. **Does not prove existence.** Uniqueness and stability do not imply that a solution exists; existence must be established separately (e.g. by constructive methods or functional analysis).

7. Conclusion

The energy method is a powerful, general technique for establishing the uniqueness and stability of solutions to Initial Boundary Value Problems for PDEs. The central idea is elegant: instead of finding the solution explicitly, one assigns an energy functional $E(t) \geq 0$ to the solution, derives an ordinary differential inequality for $E(t)$, and applies Gronwall's inequality to control the growth of E.

For the heat equation, the energy decays — diffusion is inherently dissipative. For the wave equation, the energy is conserved in the absence of forcing — reflecting the physical conservation of mechanical energy. Damping introduces dissipation and drives the energy to zero. In all three cases, the same two-step strategy yields clean, rigorous proofs of uniqueness.

The key mathematical tools required are the L^2 norm, integration by parts (which converts second derivatives to first derivatives and eliminates boundary terms), and Gronwall's inequality (which solves the resulting ODE inequality). The Poincaré inequality further allows one to obtain exponential decay rates for the heat equation.

The energy method does not require knowing the explicit solution, making it applicable even when closed-form solutions are unavailable. Together with the maximum principle [9] and existence theories, it forms one of the three pillars of the analysis of PDEs.

AI Assistance Disclosure

- **AI Tool:** Claude (Anthropic)
  Prompt: Write me a Latex code to generate the sample report and write the hand written notes and mathematical contents in the generated file.
  Usage: Used to assist in structuring the report and formatting the LaTeX source. All mathematical content, derivations, and proofs were drawn from the provided study notes and standard PDE references, and were verified independently.

- **AI Tool:** Gemini (Google)
  Prompt: write me a code using html, css and javascript to make a interactive environment to show how a reaction diffusion model (pde) works in real life. make it mathematically correct. the equation is $u_t = ku_{xx} + ru$.
  Usage: Used to generate a functional HTML/JS environment that numerically

solves a linear reaction-diffusion PDE using finite difference spatial discretization and Euler time integration.

- **AI Tool:** Gemini (Google)
  Prompt: write me a code using html, css and javascript to make a interactive environment to show how a damped wave model (pde) works in real life. make it mathematically correct. the equation is $u_{tt} + 2\gamma u_t = c^2 u_{xx}$.
  Usage: Used to write an HTML, CSS, and JavaScript script that builds an interactive, mathematically accurate simulation of a damped wave PDE model.

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